
Homework 2

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Problem 1

Problem 2

Problem 3

$$\partial A := \overline{A} \setminus \text{Int}(A).$$

Let A be closed. Then $A = \overline{A}$. Note that $\overline{A} \setminus \text{Int}(A) \subseteq \overline{A}$, and $\partial A = \overline{A} \setminus \text{Int}(A)$. Therefore, $\partial A \subseteq \overline{A} = A$.

Now for the converse, let $\partial A \subseteq A$. It follows that $\overline{A} \setminus \text{Int}(A) \subseteq A$. Note that $\overline{A} = \text{Int}(A) \cup (\overline{A} \setminus \text{Int}(A))$. Since $\text{Int}(A) \subseteq A$ and $\overline{A} \setminus \text{Int}(A) \subseteq A$, we have that their union must also be a subset of A . That is,

$$\overline{A} = \text{Int}(A) \cup (\overline{A} \setminus \text{Int}(A)) \subseteq A.$$

Since we also know for all sets that $A \subseteq \overline{A}$, we have that $A = \overline{A}$, and A is closed.

Problem 4

Problem 5

Let $f : X \rightarrow Y$ be continuous, and let Y be Hausdorff.

Problem 6

Problem 7